

PAVAN DHARMA ADAPA

Healthcare Data Analyst

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PROFESSIONAL SUMMARY

M.S. in Computer Science graduate focused on healthcare data. Analyzes public CMS claims-denial filings and CDC survey microdata in SQL and Python, with documented methods and stated limitations. As a graduate research assistant, cleaned and analyzed a 900,000+ record livestock-disease outbreak dataset for Computer Science and Veterinary Medicine faculty.

TECHNICAL SKILLS

Analysis & Reporting: SQL, Python (pandas, NumPy, scikit-learn), R, Excel, Power BI

Healthcare Data: CMS claims-denial and appeal filings, CDC survey microdata, synthetic EHR data (Synthea), quality measures (HEDIS, CMS eCQM), SNOMED CT, LOINC

Data Management: Data procurement and validation, quality control, dimensional modeling, reproducible pipelines, technical documentation, Git

PROFESSIONAL EXPERIENCE

Data Analytics Intern | *Agricultural Microbiomes Lab, Mississippi State University, Starkville, MS* Jul 2026 – Present

- Contributed **36 merged pull requests** to the lab's open-source DNA sequencing pipeline (maintained by Li Zhang), making it a documented, tested tool that installs in one command and runs on Linux, macOS, WSL2 and Windows.
- Validated pipeline output against an independently built implementation, **agreeing to within 0.01 percentage points** on every species and genus.

Graduate Teaching Assistant | *Mississippi State University, Starkville, MS* Sep 2025 – May 2026

- Taught data cleaning, transformation and visualization in Python and R to **100+ students across six courses**.

Graduate Research Assistant | *Mississippi State University, Starkville, MS* Aug 2024 – Aug 2025

- Cleaned and analyzed a **900,000+ record** Vesicular Stomatitis Virus (VSV) livestock-outbreak dataset, and presented the findings to Computer Science and Veterinary Medicine faculty.
- Built a Python/Selenium scraper that collected genome metadata for a poultry respiratory-disease project, **cutting collection from hours to under 10 minutes per dataset**.

Marketing Data Analyst Intern | *Viral Fission, Hyderabad, India* May 2023 – Dec 2023

- Built automated Power BI and Google Analytics 4 reporting for campaign performance, **halving manual reporting time** for a team running national brand campaigns.

HEALTHCARE ANALYTICS PROJECTS

Claims Denials & Appeals — *CMS Transparency in Coverage, 2021–2023*

- Built a **15,545 plan-year** SQL panel covering **126.9M denied claims** across every federal Marketplace plan, using window functions over a two-grain fact design.
- Found only **0.168% of denials are ever appealed while 40.1% of appeals are overturned**, consistent across three years of issuer-reported filings.

Clinical Quality Measures Engine — *HEDIS / CMS eCQM · synthetic non-PHI EHR (Synthea)*

- Computed HEDIS and CMS eCQM measures over a **13,810-patient EHR** (820,993 encounters, 10.8M observations), holding each measure as a versioned specification with its SNOMED CT and LOINC value sets separate from the engine — so an annual steward update is a reviewable diff a clinical analyst signs off, not a code change.
- Traced **203 of 204** diabetes-measure failures to missing HbA1c tests, an artifact of how the Synthea generator schedules labs, and reported it as an engine check rather than a finding about care.

EDUCATION

M.S., Computer Science — Mississippi State University, Mississippi State, MS May 2026

B.Tech, Computer Science — Malla Reddy University, Hyderabad, India May 2024

Publication (under revision): Li Zhang, Saikanth Ratnavale, Pavan Dharma Adapa, Sai Harika Gade, Michael E. Navicky — Computational Tools for Modeling and Predicting Respiratory Disease Spread in Commercial Poultry Production.